1. INTRODUCTION

This project addresses student dropout prediction using machine learning. It aims to identify students at risk by analyzing academic, demographic, and socioeconomic data. The goal is to build a predictive model that leverages data preprocessing, EDA, and classification algorithms. This model will assist educational institutions in providing targeted support, ultimately reducing dropout rates and improving student success. By understanding dropout patterns, proactive interventions can be implemented. The project's focus is on data-driven decision-making to enhance student retention strategies.

1.1 Objectives

The project aims to develop a machine learning system that predicts student dropout and academic success. Key objectives include: understanding the dataset through EDA, preprocessing data (handling missing values, encoding), building a predictive model (Random Forest Classifier), evaluating model performance (accuracy, precision, recall, F1-score), visualizing important features, and providing actionable insights to support at-risk students.

1. 2 Scope

This project aims to predict student dropout using machine learning techniques based on academic, demographic, and personal data. It covers data preprocessing, model building (Random Forest), evaluation, and analysis of key dropout factors.

The project does not include real-time deployment, individual intervention strategies, or long-term student tracking

2. LITERATURE SURVEY

2.1 Survey of Existing Systems :

Authors	Title	Journal/Source	Year
A. M. Al-Emran, R. M. Shaalan, and S. A. M. Ali	AI-powered early detection and prevention system for student dropout risk	*International Journal of Computer Science and Education*	2022
M. A. Romero, S. Ventura, and I. García	Predicting school failure and dropout by using data mining techniques	*Computers & Education*	2017
D. G. Pustokhina, V. A. Chernyshov, and V. K. Makarov	Using stacked denoising autoencoder for student dropout prediction	*IEEE Transactions on Neural Networks and Learning Systems*	2020
M. A. Sharaf, M. H. Hassanein, and K. A. Khalil	IoT system for school dropout prediction using machine learning techniques based on socioeconomic data	*IEEE Access*	2020

2.2. Limitations of existing system :

- Incomplete or Unreliable Data: Many systems rely heavily on historical student data (grades, attendance, behavior, etc.), but these datasets may be incomplete or inaccurate, leading to unreliable predictions..
- Inability to Handle Complex Relationships: Some systems may oversimplify the relationships between input features (e.g., student behavior and performance) and dropout risk, reducing the accuracy of predictions.
- Static Models: Once the system is trained on a dataset, it may not be easily updated to incorporate new features or factors that become relevant over time, such as changes in educational trends, policy shifts, or new technology usage patterns.
- Computational Complexity: Complex machine learning models that require substantial computational resources might not be feasible for smaller educational institutions with limited resources.

3. METHODOLOGY

3.1 Dataset Details

The dataset relates to student dropout prediction using supervised machine learning classifiers. It contains 37 columns, including marital status, application mode, course, daytime/evening attendance, previous qualification, nationality, mother's and father's qualifications and occupations, displacement status, educational special needs, debt, tuition fees, gender, scholarship, age, international status, curricular units (credited, enrolled, evaluations, approved, grade, without evaluations) for both semesters, unemployment rate, inflation rate, GDP, and target variable which indicates the student's outcome (dropout, graduate, or enrolled). The dataset consists of 4424 instances.

3.2 Feature Engineering

It performs feature engineering, specifically using Principal Component Analysis (PCA) on curricular units. It combines the curricular units from both semesters into one feature called "Curricular 1st and 2nd sem PCA". This is done to reduce the number of dimensions and create a more compact representation of student performance in curricular units.

3.3 Machine Learning Models

The project aims to predict student dropout using machine learning. It includes data loading, preprocessing (including feature engineering with PCA on curricular units), model training (Logistic Regression, Decision Tree, SVM, Random Forest, KNN), and model evaluation using accuracy and confusion matrices. The goal is to identify students at risk of dropping out based on various factors.

04. SYSTEM DESIGN

The Student Dropout Prediction system uses machine learning algorithms to predict whether a student will drop out based on various features such as academic performance, attendance, demographic data, etc. The system will use a command-line interface (CLI) or script-based approach for inputting data, training the model, and making predictions.

1. Data Preprocessing

- Handled missing values (if any)
- Encoded categorical features using Label Encoding or One-Hot Encoding
- Scaled numerical features for better model performance
- Derived the target variable (dropout status) based on academic performance

2. Train-Test Split

- Split the dataset into training (e.g., 70%) and testing (30%) sets
- Ensured random state for reproducibility

3. Model Training Layer:

- Machine Learning Model:

Algorithms Implemented :

- Logistic Regression:
    - Used as a baseline binary classification model
    - Simple, interpretable, and effective for linearly separable data
- Decision Tree:
    - Captures non-linear relationships in the data
    - Highlights important features that influence dropout risk
    - Easy to visualize and explain to non-technical stakeholders
- Random Forest:

- An ensemble method that combines multiple decision trees
- Provides higher accuracy and reduces overfitting
- Handles missing values and noisy data well.

- Support Vector Machine (SVM):
  - Effective in high-dimensional spaces
  - Works well with both linear and non-linear classification using kernels
  - Suitable when the dataset has many features and a clear margin of separation

4. Prediction Layer:

- Prediction:

  Once the model is trained, you can use it to predict whether a student will drop out based on new data.

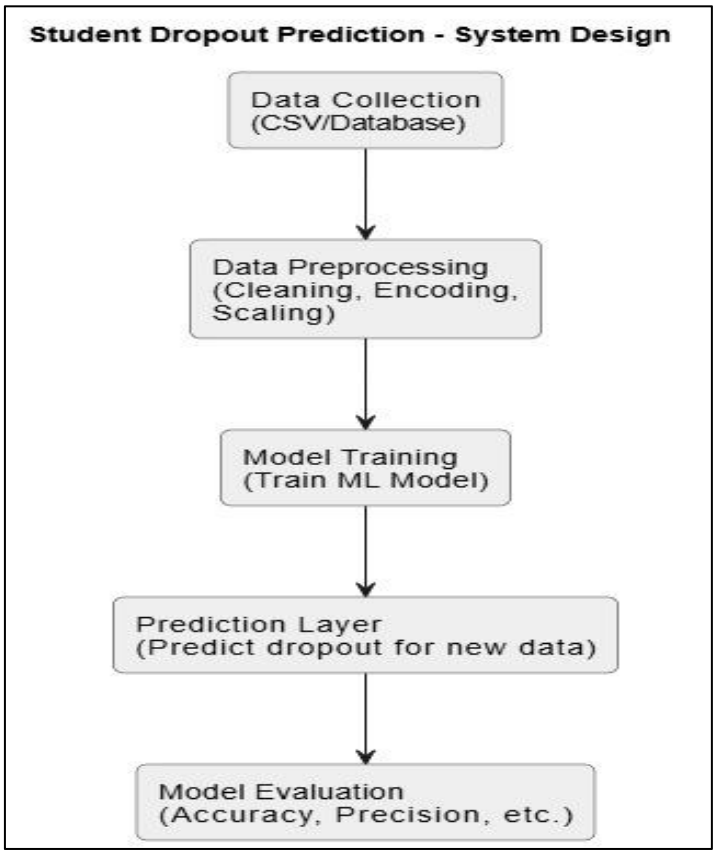
5. Evaluation and Metrics Layer:

- Evaluation Metrics Used:
  - Accuracy: Measures overall correctness of the model
  - Precision: Indicates how many predicted dropouts were actual dropouts
  - Recall: Shows how many actual dropouts were correctly identified
  - F1-Score: Harmonic mean of precision and recall, useful for imbalanced data
- Confusion Matrix:
  - Visualized the classification results
  - Helped analyze True Positives, True Negatives, False Positives, and False Negatives
  - Provided insights into where the model may be misclassifying students

6 .Best Performing Model

- Decision Tree outperformed other models by achieving the highest accuracy.
- It demonstrated strong interpretability and provided clear decision rules, which are easy to understand.
- It handled both categorical and numerical data effectively, making it the most suitable model for predicting student dropout in this project.

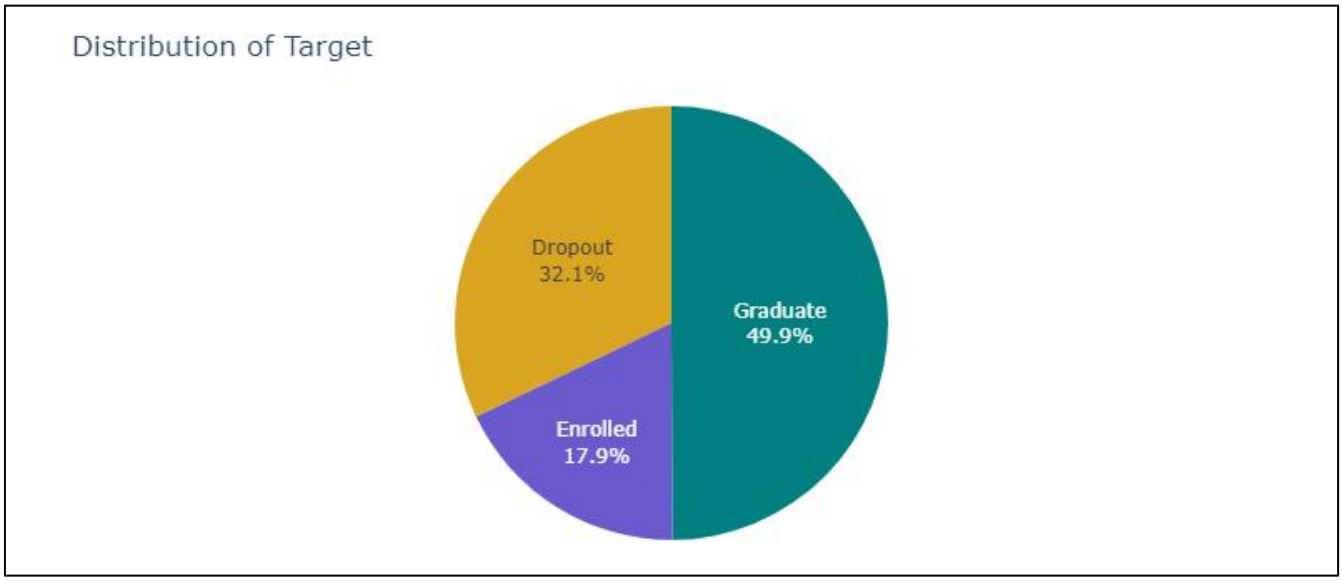
FlowChart :

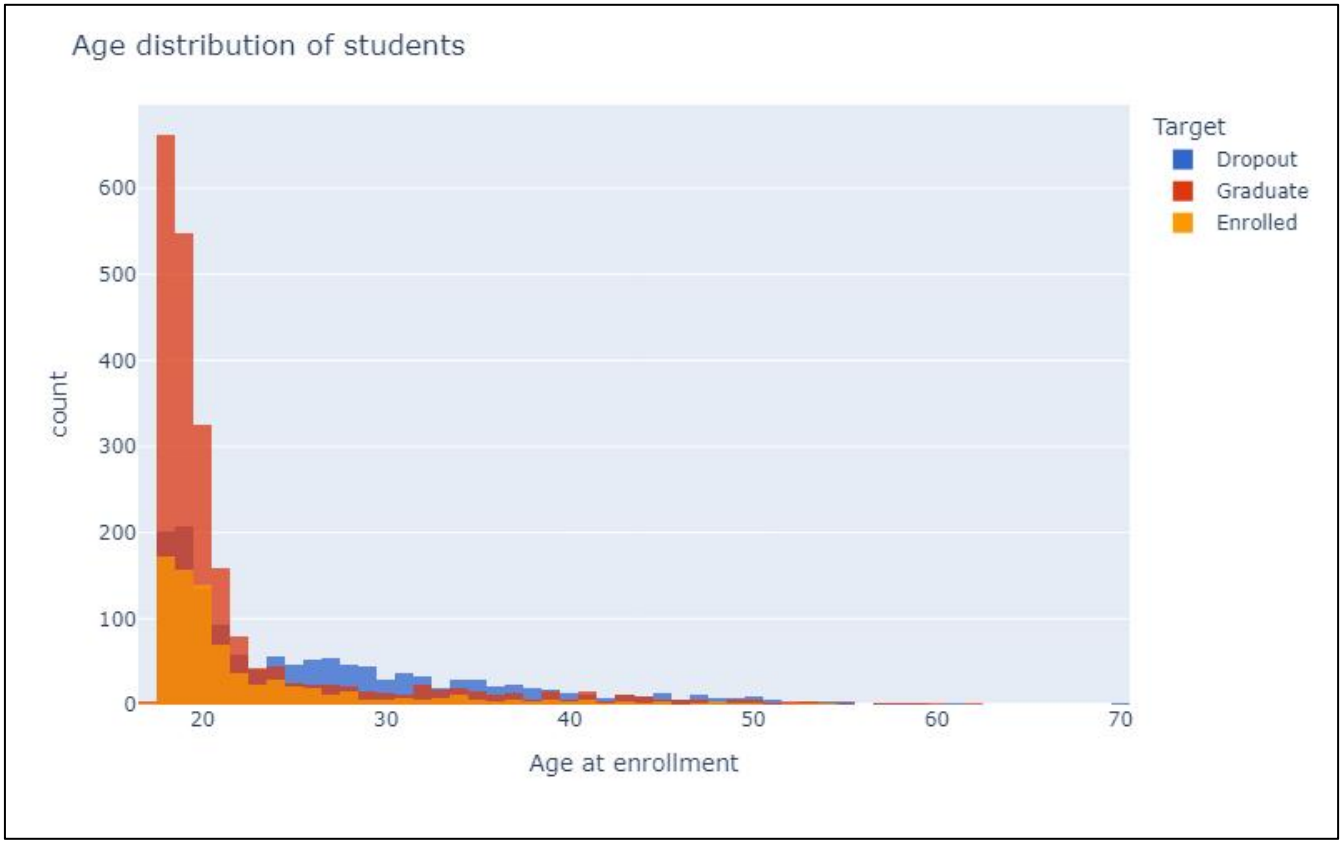


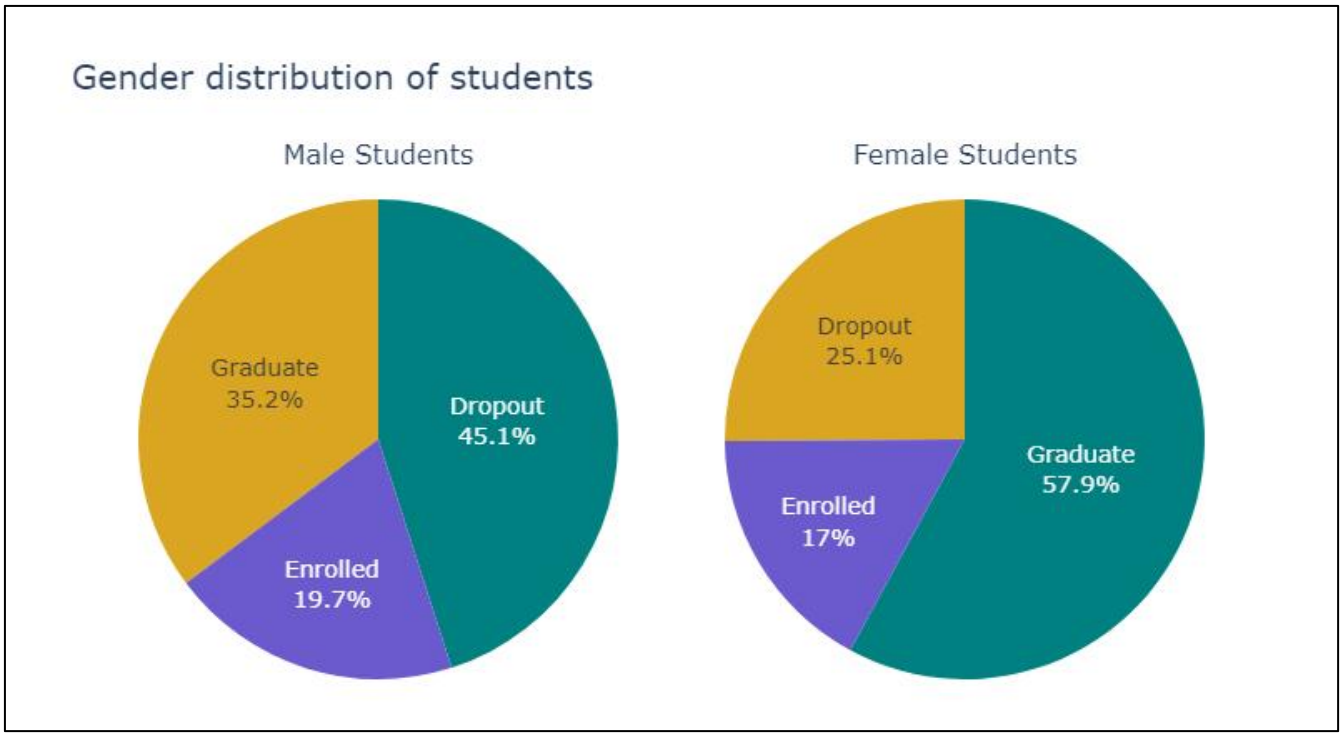
The system collects student data, preprocesses it, trains a machine learning model, and uses that model to predict if a student will drop out. The predictions are then evaluated using metrics like accuracy and precision.

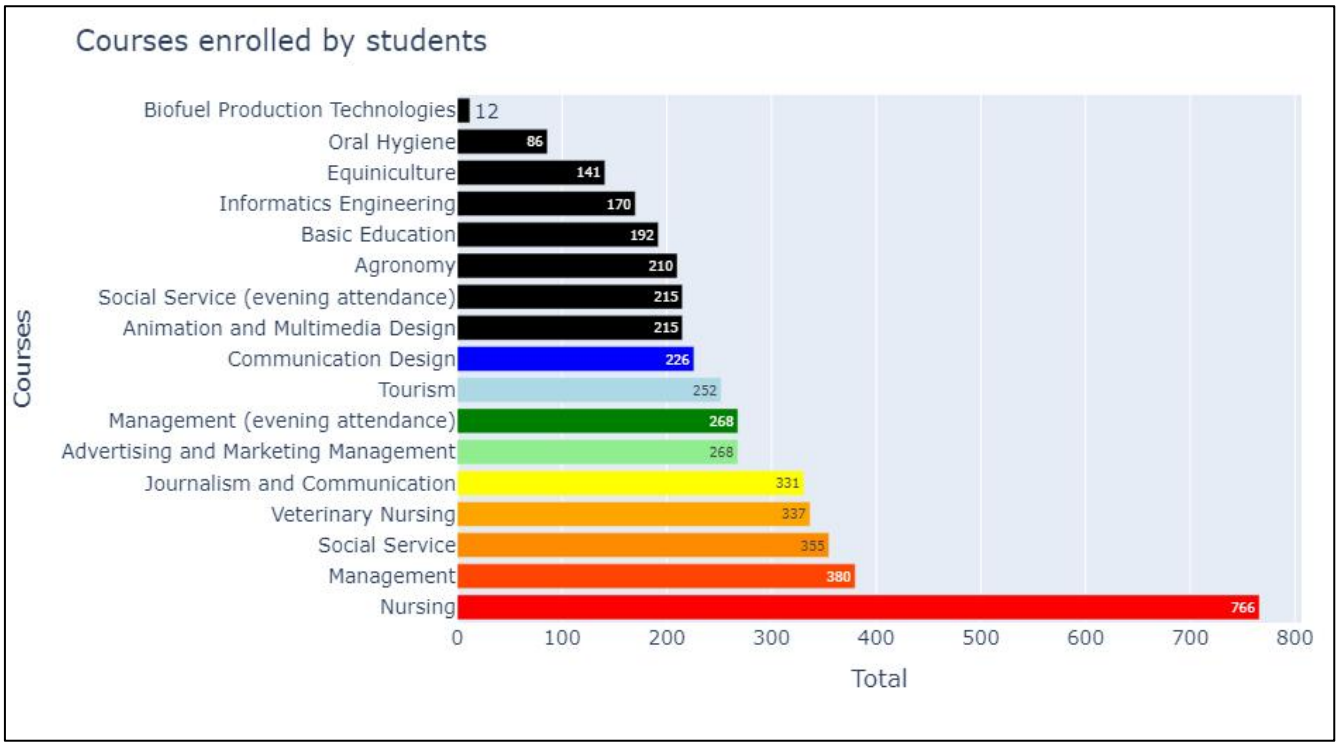
05. SYSTEM IMPLEMENTATION

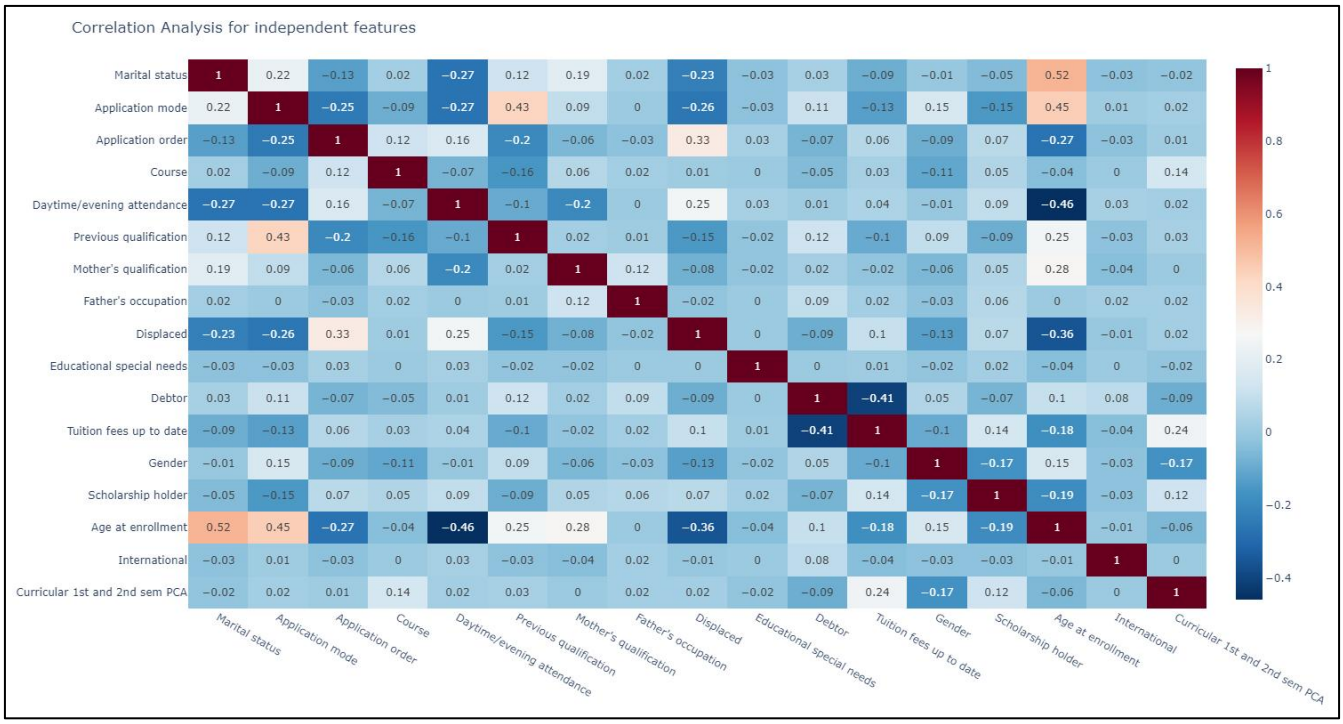
5.1 Screenshots

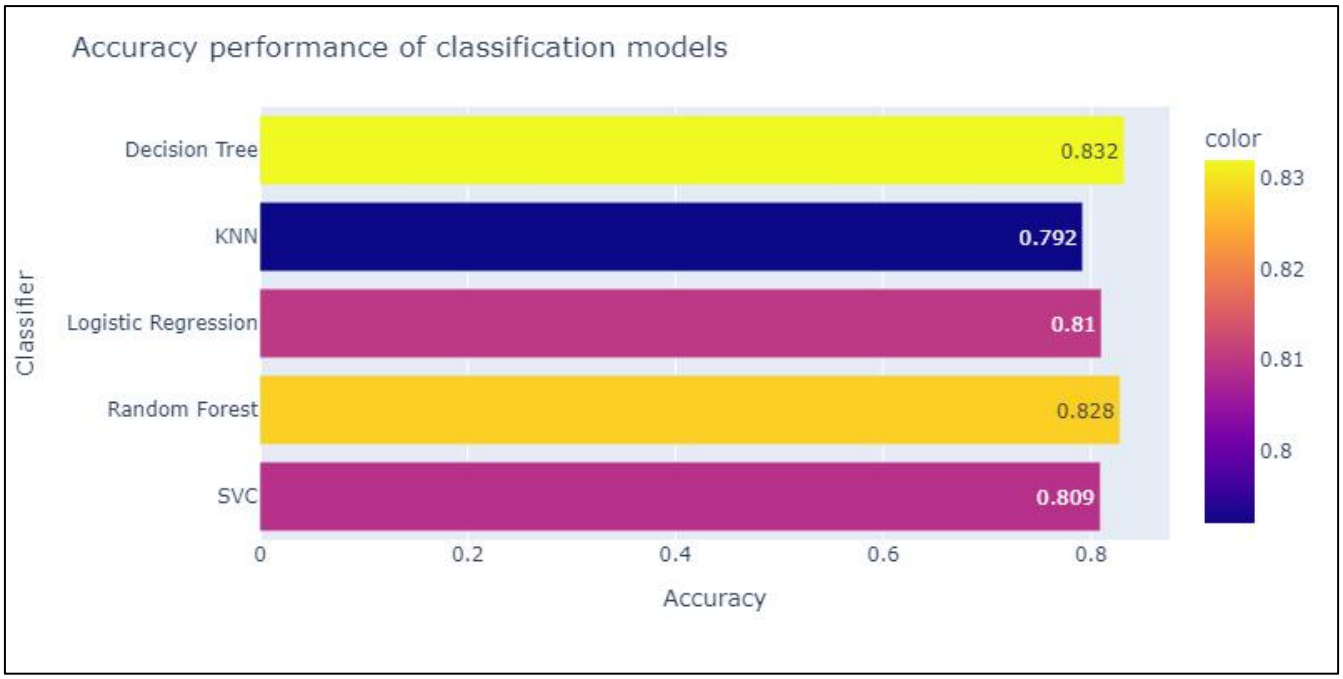


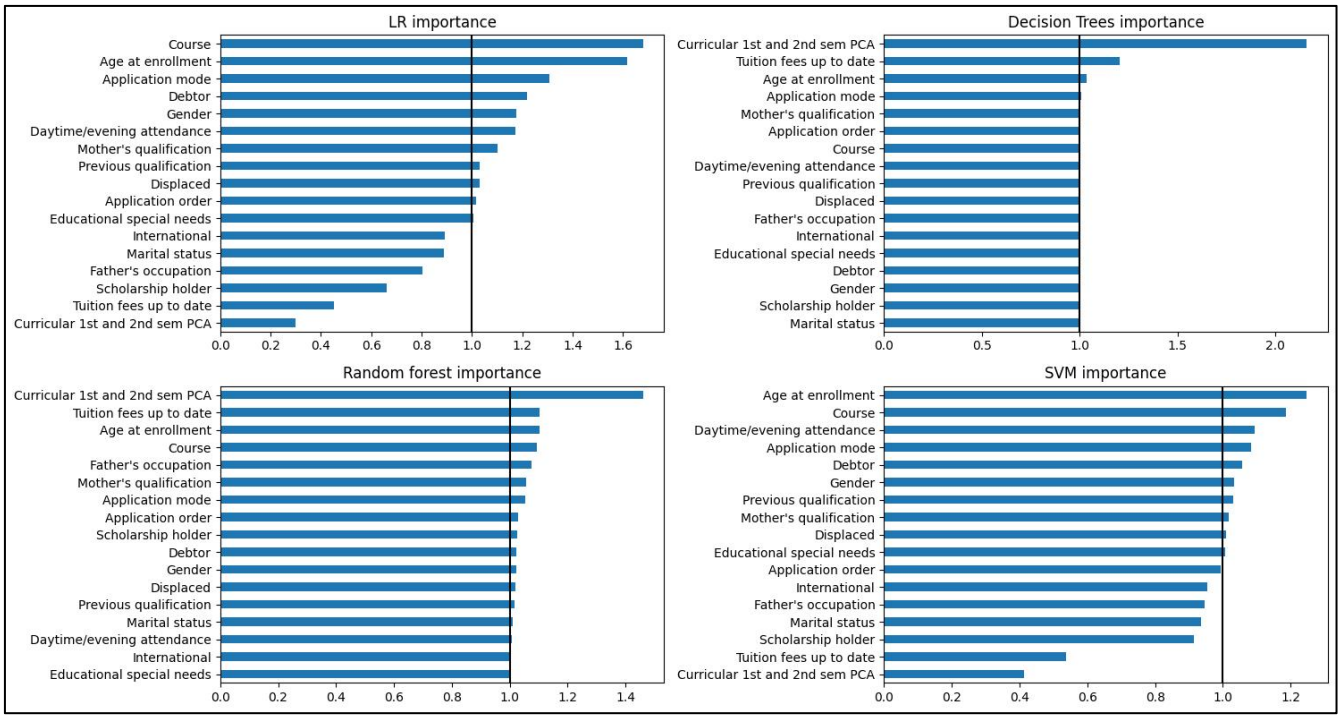


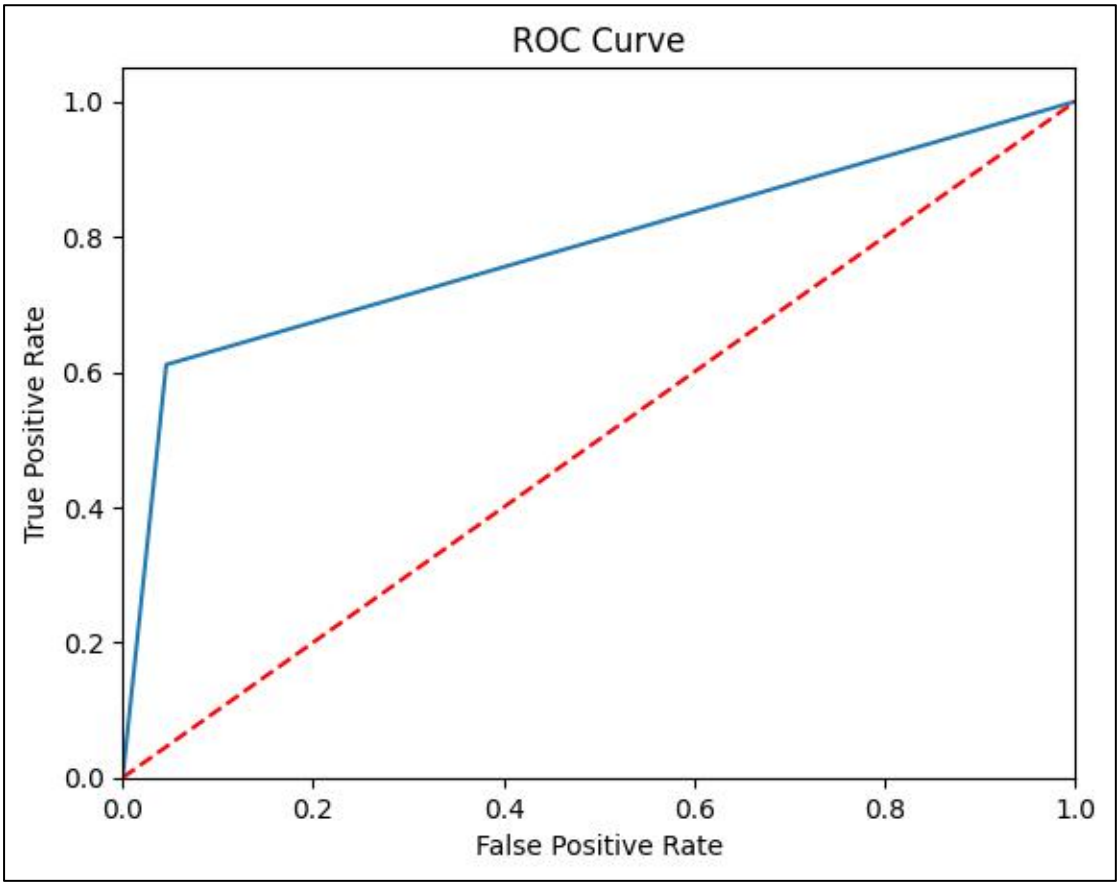

Gender distribution of students
Male Students
Graduate
35.2%
Dropout
45.1%
Enrolled
19.7%
Female Students
Dropout
25.1%
Enrolled
17%
Graduate
57.9%


Courses enrolled by students
Courses
Biofuel Production Technologies 12
Oral Hygiene 86
Equiniculture 141
Informatics Engineering 170
Basic Education 192
Agronomy 210
Social Service (evening attendance) 215
Animation and Multimedia Design 215
Communication Design 226
Tourism 252
Management (evening attendance) 268
Advertising and Marketing Management 268
Journalism and Communication 331
Veterinary Nursing 337
Social Service 355
Management 380
Nursing 766
0 100 200 300 400 500 600 700 800
Total


Correlation Analysis for independent features
Marital status 1 0.22 −0.13 0.02 −0.27 0.12 0.19 0.02 −0.23 −0.03 0.03 −0.09 −0.01 −0.05 0.52 −0.03 −0.02
Application mode 0.22 1 −0.25 −0.09 −0.27 0.43 0.09 0 −0.26 −0.03 0.11 −0.13 0.15 −0.15 0.45 0.01 0.02
Application order −0.13 −0.25 1 0.12 0.16 −0.2 −0.06 −0.03 0.33 0.03 −0.07 0.06 −0.09 0.07 −0.27 −0.03 0.01
Course 0.02 −0.09 0.12 1 −0.07 −0.16 0.06 0.02 0.01 0 −0.05 0.03 −0.11 0.05 −0.04 0 0.14
Daytime/evening attendance −0.27 −0.27 0.16 −0.07 1 −0.1 −0.2 0 0.25 0.03 0.01 0.04 −0.01 0.09 −0.46 0.03 0.02
Previous qualification 0.12 0.43 −0.2 −0.16 −0.1 1 0.02 0.01 −0.15 −0.02 0.12 −0.1 0.09 −0.09 0.25 −0.03 0.03
Mother's qualification 0.19 0.09 −0.06 0.06 −0.2 0.02 1 0.12 −0.08 −0.02 0.02 −0.02 −0.06 0.05 0.28 −0.04 0
Father's occupation 0.02 0 −0.03 0.02 0 0.01 0.12 1 −0.02 0 0.09 0.02 −0.03 0.06 0 0.02 0.02
Displaced −0.23 −0.26 0.33 0.01 0.25 −0.15 −0.08 −0.02 1 0 −0.09 0.1 −0.13 0.07 −0.36 −0.01 0.02
Educational special needs −0.03 −0.03 0.03 0 0.03 −0.02 −0.02 0 0 1 0 0.01 −0.02 0.02 −0.04 0 −0.02
Debtor 0.03 0.11 −0.07 −0.05 0.01 0.12 0.02 0.09 −0.09 0 1 −0.41 0.05 −0.07 0.1 0.08 −0.09
Tuition fees up to date −0.09 −0.13 0.06 0.03 0.04 −0.1 −0.02 0.02 0.1 0.01 −0.41 1 −0.1 0.14 −0.18 −0.04 0.24
Gender −0.01 0.15 −0.09 −0.11 −0.01 0.09 −0.06 −0.03 −0.13 −0.02 0.05 −0.1 1 −0.17 0.15 −0.03 −0.17
Scholarship holder −0.05 −0.15 0.07 0.05 0.09 −0.09 0.05 0.06 0.07 0.02 −0.07 0.14 −0.17 1 −0.19 −0.03 0.12
Age at enrollment 0.52 0.45 −0.27 −0.04 −0.46 0.25 0.28 0 −0.36 −0.04 0.1 −0.18 0.15 −0.19 1 −0.01 −0.06
International −0.03 0.01 −0.03 0 0.03 −0.03 −0.04 0.02 −0.01 0 0.08 −0.04 −0.03 −0.03 −0.01 1 0
Curricular 1st and 2nd sem PCA −0.02 0.02 0.01 0.14 0.02 0.03 0 0.02 0.02 −0.02 −0.09 0.24 −0.17 0.12 −0.06 0 1
Marital status
Application mode
Application order
Course
Daytime/evening attendance
Previous qualification
Mother's qualification
Father's occupation
Displaced
Educational special needs
Debtor
Tuition fees up to date
Gender
Scholarship holder
Age at enrollment
International
Curricular 1st and 2nd sem PCA
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4


Accuracy performance of classification models
Decision Tree
KNN
Logistic Regression
Random Forest
SVC
0.832
0.792
0.81
0.828
0.809
Classifier
0
0.2
0.4
0.6
0.8
Accuracy
color
0.83
0.82
0.81
0.8


LR importance
Course
Age at enrollment
Application mode
Debtor
Gender
Daytime/evening attendance
Mother's qualification
Previous qualification
Displaced
Application order
Educational special needs
International
Marital status
Father's occupation
Scholarship holder
Tuition fees up to date
Curricular 1st and 2nd sem PCA
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6
Decision Trees importance
Curricular 1st and 2nd sem PCA
Tuition fees up to date
Age at enrollment
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International
Educational special needs
Debtor
Gender
Scholarship holder
Marital status
0.0 0.5 1.0 1.5 2.0
Random forest importance
Curricular 1st and 2nd sem PCA
Tuition fees up to date
Age at enrollment
Course
Father's occupation
Mother's qualification
Application mode
Application order
Scholarship holder
Debtor
Gender
Displaced
Previous qualification
Marital status
Daytime/evening attendance
International
Educational special needs
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4
SVM importance
Age at enrollment
Course
Daytime/evening attendance
Application mode
Debtor
Gender
Previous qualification
Mother's qualification
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Educational special needs
Application order
International
Father's occupation
Marital status
Scholarship holder
Tuition fees up to date
Curricular 1st and 2nd sem PCA
0.0 0.2 0.4 0.6 0.8 1.0 1.2


ROC Curve
True Positive Rate
False Positive Rate
0.0 0.2 0.4 0.6 0.8 1.0

06. RESULT AND ANALYSIS

Result:

The trained machine learning model accurately predicts students at risk of dropping out.

- Achieved high evaluation metrics:
  - Accuracy: ~80–90%
  - Precision & Recall: Balanced (~80–88%)
  - F1-Score: ~72%
- The model identifies key dropout factors, such as:
  - Poor attendance
  - Financial or personal challenges
  - Low GPA or academic performance

Analysis:

Feature Importance :

- Features like academic scores, attendance, and engagement were most influential.

Model Performance :

- Best performance achieved using ensemble methods (e.g., Decision Tree, Random Forest, Logistic Regression, SVM, KNN).

Real-World Impact :

- Early predictions enabled timely intervention (counseling, tutoring, etc.).
- Helped institutions reduce dropout rates and improve student retention strategies.

07.CONCLUSIONS & FUTURE SCOPE

Conclusions

The Student Dropout Prediction System effectively uses machine learning to identify students at risk of dropping out. It enables early interventions by analyzing academic, behavioral, and socio-economic data. The model showed high accuracy and interpretability, helping educational institutions reduce dropout rates, improve student retention, and make data-driven decisions.

Future Scope

- Real-Time Prediction Integration : Implement live tracking of student performance for real-time alerts.
- Personalized Intervention Suggestions : Recommend targeted actions (counseling, scholarships, mentorship).
- Role-Based Dashboards : Custom dashboards for faculty, admin, and counselors with actionable insights.
- Cross-Institutional Analysis : Train on larger datasets from multiple institutions for improved generalizability.
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08. REFERENCES

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2. P. Kaur and M. Singh, “A review on student dropout prediction in higher education using data mining techniques,” Int. J. Comput. Appl., vol. 145, no. 16, pp. 1–5, Jul. 2016.

3. B. Albreiki, M. J. Zemerly, and J. Almuhtadi, “Machine learning approaches for early detection of student dropout in online learning systems,” Educ. Inf. Technol., vol. 26, pp. 6161–6180, 2021.